

Chapter 51 — Pathophysiology of Urinary Tract Obstruction — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Posterior urethral valves (PUV) + urethral atresia** ≈ 50% of lower urinary tract obstruction (LUTO)
- **Ureteropelvic junction obstruction (UPJO)** = commonest upper tract cause; 1 in 1500, male 2:1
- **Second-trimester oligo/anhydramnios** in LUTO → 60–80% mortality
- **Anteroposterior diameter (APD)** ≥ 7 mm (16–27 wk), ≥ 10 mm (≥ 28 wk) → consider obstruction
- **Favourable fetal urine:** Na < 100, Cl < 90 mEq/L, osm < 200 mOsm/L
- **Voiding cystourethrogram (VCUG)** = gold standard for vesicoureteral reflux (VUR)
- **Mercaptoacetyltriglycine (MAG3)** = drainage and level; **dimercaptosuccinic acid (DMSA)** = scar
- **Bladder > 40 cm H₂O** = hostile, threatens kidneys
- **Postobstructive diuresis (POD):** ≈ 2% unilateral vs ≈ 50% bilateral
- **Pyeloplasty:** re-anastomose **anterior** to crossing vessel

PATHOPHYSIOLOGY OF PAEDIATRIC OBSTRUCTION

- **Incidence:** ≈ 1 in 500 live births
- **LUTO:** 2–3 per 10,000 fetuses; 1 in 5000–25,000 pregnancies
- **High-grade obstruction:** lifetime end-stage renal disease (ESRD) risk up to 30%
- **Governing risk:** permanent damage to developing kidney
- **Axes:** acute/chronic · partial/complete · intrinsic/extrinsic · uni/bilateral

CLINICAL PRESENTATIONS

- **Prenatal detection:** earliest 11–14 wk; most 18–20 wk
- **Commonest prenatal finding:** hydronephrosis
- **Prenatal US signs:** megacystis, keyhole, echogenic/cystic kidneys, thick bladder, ascites, oligohydramnios
- **Keyhole sign** = dilated posterior urethra → LUTO
- **Other LUTO causes:** anterior valves, megalourethra, prolapsing ureterocele, cloaca, megacystis–microcolon–intestinal hypoperistalsis syndrome (MMIHS), trisomy 18/21
- **LUTO kills via** ESRD + pulmonary hypoplasia
- **Extrinsic causes:** crossing vessel, retroperitoneal fibrosis, nodes, pelvic/spinal tumour, faecal impaction
- **Intraluminal causes:** calculi, sloughed papillae, clot, upper tract bleeding
- **Other upper tract causes:** ureterovesical junction obstruction, ureterocele, ectopic ureter, fibroepithelial polyp, calyceal diverticulum
- **Functional lower tract cause:** neurogenic bladder, myogenic failure
- **Lower tract tumour:** prostatic rhabdomyosarcoma

DEFINITION OF OBSTRUCTION

- **Partial** → may resolve; **high-grade** → intervene
- **Prenatal obstruction** → pulmonary hypoplasia, limb deformity

DIAGNOSIS

- **Central task:** surgery-needing vs spontaneously resolving
- **CAKUT** (congenital anomalies of kidney and urinary tract): 1 in 250–1000 pregnancies
- **Fetal kidneys** 9 wk; **bladder** 12 wk; **absent at 15 wk** = abnormal
- **Urinary tract dilation (UTD) system:** 8 societies; split at 28 wk
- **Most reliable prenatal parameter:** APD (ERKNet consensus, 2022)
- **Below APD threshold** → obstruction only if megacystis or low fluid
- **Primary prenatal modality — ultrasound (US):** cost-effective, part of anomaly screening

- **Fetal MRI:** adjunct — oligohydramnios, maternal obesity, pre-intervention
- **Megacystis:** fails to empty over 40 min at ≥ 18 wk
- **Fetal urine also:** Ca < 8 mg/dL, β₂-microglobulin < 6 mg/L
- **Vesicocentesis:** 3 serial samples; **last** = current function
- **Fetal urine uses:** obstructive vs non-obstructive; monitor LUTO intervention
- **12PUV peptide signature:** ESRD < 2 yr; sensitivity 88%, specificity 95%
- **Urinalysis:** initial study; low specific gravity = concentrating defect
- **Fractional excretion of sodium (FENa):** < 1% prerenal; > 1% tubular damage/acute tubular necrosis
- **FENa in obstruction:** low acutely; high once tubules damaged
- **FENa** = (urine Na × serum creatinine) / (urine creatinine × serum Na) × 100
- **Best paediatric glomerular filtration rate (GFR) estimate:** cystatin C
- **Postnatal biomarkers:** neutrophil gelatinase-associated lipocalin, kidney injury molecule-1, cystatin C, transforming growth factor (TGF)-β1, retinol-binding protein → expedite surgery
- **Imaging tiers:** primary US, scintigraphy, VCUG, MR urography → secondary retrograde/antegrade pyelography → tertiary intravenous pyelography (IVP), CT urography (CTU), Whitaker
- **First-line postnatal imaging:** US
- **Technetium-99m DTPA:** function + perfusion only, no drainage
- **PUV on VCUG:** dilated posterior urethra, voiding phase
- **Caecoureterocele:** filling defect at trigone
- **VUR grades** 1–5
- **High-grade VUR** → secondary UPJO
- **MR urography:** anatomy + function, no radiation; ectopic ureter
- **Retrograde** = cystoscopic cannulation; **antegrade** = percutaneous
- **IVP:** seldom used; bowel prep improves accuracy
- **CTU:** preferred over IVP despite **higher** radiation
- **Whitaker indications:** equivocal tests, poor function, intermittent, gross dilation + positive renogram
- **Whitaker obstructed:** > 12 cm H₂O or continuous rise (source); classical bands < 15 / 15–22 / > 22 cm H₂O = unobstructed / equivocal / obstructed
- **Uroflowmetry:** voided volume, maximum flow rate, curve

PATHOLOGICAL CHANGES

- **Bladder remodelling:** hypertrophy → compensation → decompensation

- **Compliance** = extracellular matrix (ECM); interleukin-6, tumour necrosis factor- α , monocyte chemoattractant protein-1 $\rightarrow$ collagen
- **Immature kidney**: apoptosis, impaired nephrogenesis
- **Mature kidney**: proliferation $\rightarrow$ fibrosis
- **Fibrosis marker**: α -smooth muscle actin (α -SMA)
- **Master fibrotic pathway**: TGF- β $\rightarrow$ Smad
- **miR-34a** $\rightarrow$ epithelial-to-mesenchymal transition
- **Worse disease**: bilateral obstruction or solitary kidney
- **Ischaemic injury** within 2 h of complete obstruction
- **Early wasting**: Na, K, HCO₃, Mg, Ca, PO₄
- **Late retention**: K, H⁺, Cl, NH₄⁺
- **Acid-base**: distal renal tubular acidosis
- **Aquaporins 1–4 + Na transporters** downregulated

CLINICAL IMPACT

- **Renal parenchymal disease** = 60–80% of paediatric secondary hypertension
- **Hypertension**: $\approx$ 77% bilateral vs $\approx$ 20% unilateral
- **Unilateral**: renin lateralises to affected kidney
- **Compensatory growth**: insulin-like growth factor-1, TGF- β 1, hepatocyte growth factor; glomerulomegaly
- **End-stage renal failure**: days to months of obstruction
- **Acute-on-chronic**: obstruction on chronic kidney disease (CKD) $\rightarrow$ rapid acute kidney injury
- **Recovery**: often incomplete after relief
- **POD mechanism**: aquaporin loss, medullary washout, transporter dysfunction

TREATMENT

- **Shunt/fetal cystoscopy**: LUTO with preserved function
- **Amnioinfusion**: renal agenesis or LUTO with renal failure; serial
- **Pain sources**: baroreceptors, stent, infection, decompression haematuria
- **Avoid non-steroidal anti-inflammatory drugs** if renal function compromised

- **Timed voiding**: every 1–3 h; double void for high residual
- **Continuous overnight catheter drainage**: post-PUV nocturnal polyuria; delays renal progression
- **Suprapubic tube**: urethral route impossible; short-term
- **Vesicostomy**: infant complex LUTO, unstable, unable to perform clean intermittent catheterisation; temporising
- **Surgery for LUTO**: valve excision, stricture repair, outlet reconstruction
- **Stent**: best for extrinsic ureteral obstruction
- **Nephrostomy**: needs some hydronephrosis
- **Pyeloplasty**: correct **all** intrinsic + extrinsic causes
- **Paediatric reimplant**: non-refluxing
- **CKD/ESRD**: blood pressure, diet, drugs $\rightarrow$ dialysis or transplant
- **POD duration**: resolves $<$ 24 h; rarely $>$ 48 h
- **POD risk after valve ablation**: oligohydramnios, prematurity, high creatinine

CLASSIC TRAPS

- **Trap**: dilation = obstruction — **Actually**: partial may resolve spontaneously
- **Trap**: detrusor hypertrophy is lower tract only — **Actually**: causes secondary ureterovesical junction obstruction
- **Trap**: crossing vessel is intrinsic — **Actually**: extrinsic cause of UPJO
- **Trap**: first fetal urine sample predicts function — **Actually**: the third does
- **Trap**: visible fetal ureter is normal — **Actually**: always dilated
- **Trap**: CTU gives less radiation than IVP — **Actually**: more
- **Trap**: compensatory growth = new nephrons — **Actually**: glomerulomegaly
- **Trap**: overnight drainage replaces catheterisation/augmentation — **Actually**: adjunct only
- **Trap**: amnioinfusion relieves obstruction — **Actually**: replaces fluid only
- **Trap**: generous fluid replacement in POD — **Actually**: overcorrection perpetuates it